

Table of Contents

1. Before the Workshop: Required Steps for Windows	1
2. Final Checklist	2
3. Detailed Installation Guidelines	2
<i>Install WSL 2 (Windows Subsystem for Linux)</i>	<i>2</i>
<i>Install Docker Desktop</i>	<i>4</i>
<i>Install Python</i>	<i>7</i>
<i>Install Git</i>	<i>7</i>
<i>Create A HydroShare Account</i>	<i>7</i>
<i>Install VS Code</i>	<i>7</i>
<i>Install Pixi</i>	<i>10</i>
<i>Install the QGIS</i>	<i>11</i>
4. During the Workshop: Guided Workflow Steps	12

1. Before the Workshop: Required Steps for Windows

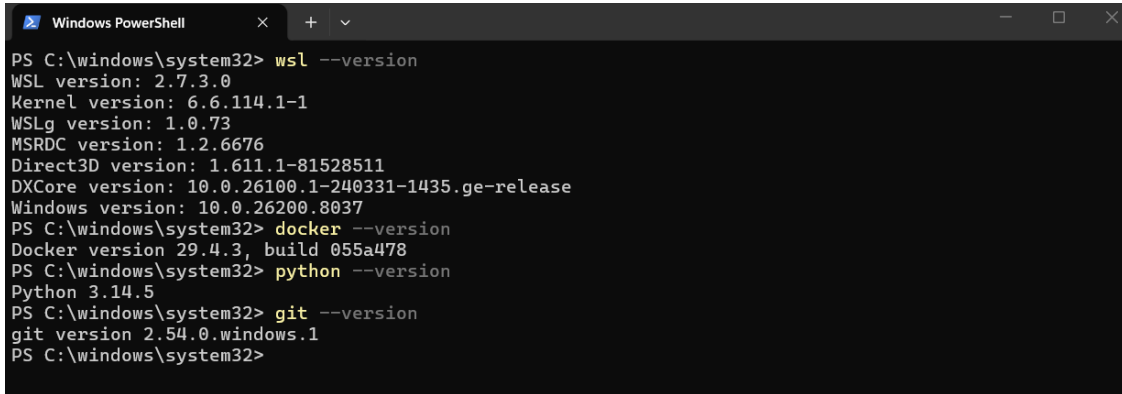
If you are using Windows, follow the instruction below to ensure required softwares are installed. After installing the required software, all workshop commands **should be run inside the Ubuntu WSL terminal**, NOT Windows native PowerShell.

- **Mandatory**
 - ✓ WSL 2 (Windows Subsystem for Linux)
 - ✓ Docker Desktop
 - ✓ Python
 - ✓ Git
 - ✓ HydroShare Account
- **Optional, but recommended**
 - ✓ VS Code
 - ✓ Install Pixi
 - ✓ QGIS

2. Final Checklist

Confirm the following commands work in your terminal and ensure that you have created a HydroShare account.

- Open a PowerShell terminal and run these commands:



```
Windows PowerShell
PS C:\windows\system32> wsl --version
WSL version: 2.7.3.0
Kernel version: 6.6.114.1-1
WSLg version: 1.0.73
MSRDC version: 1.2.6676
Direct3D version: 1.611.1-81528511
DXCore version: 10.0.26100.1-240331-1435.ge-release
Windows version: 10.0.26200.8037
PS C:\windows\system32> docker --version
Docker version 29.4.3, build 055a478
PS C:\windows\system32> python --version
Python 3.14.5
PS C:\windows\system32> git --version
git version 2.54.0.windows.1
PS C:\windows\system32>
```

- Open VS Code with WSL Ubuntu, as described in details in the following sections, and run these:



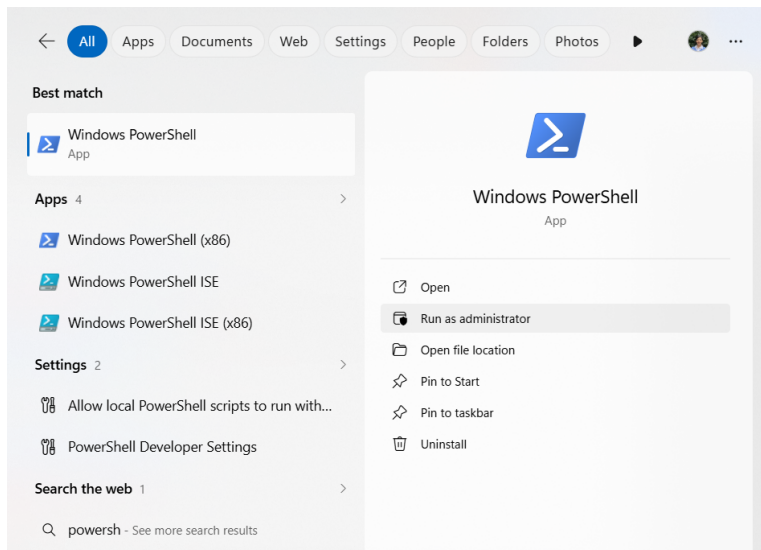
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
bash - igarousi
• igarousi@igarousidell:~$ pixi --version
pixi 0.69.0
• igarousi@igarousidell:~$ docker --version
Docker version 29.4.3, build 055a478
○ igarousi@igarousidell:~$
```

3. Detailed Installation Guidelines

Follow the instructions below for each part. It is suggested that you install these in the same order.

Install WSL 2 (Windows Subsystem for Linux)

Open a **PowerShell as Administrator** and run **wsl --install**



```
Administrator: Windows PowerShell
PS C:\windows\system32> wsl --install
Installing Windows optional component: VirtualMachinePlatform

Deployment Image Servicing and Management tool
Version: 10.0.26100.5074

Image Version: 10.0.26200.8037

Enabling feature(s)
[=====100.0%=====]
The operation completed successfully.
The requested operation is successful. Changes will not be effective until the system is rebooted.
PS C:\windows\system32>
```

Then restart the computer. After restart, check **wsl --version**

```
Administrator: Windows PowerShell
PS C:\windows\system32> wsl --version
WSL version: 2.7.3.0
Kernel version: 6.6.114.1-1
WSLg version: 1.0.73
WSRDC version: 1.2.6676
Direct3D version: 1.611.1-81528511
DXCore version: 10.0.26100.1-240331-1435.ge-release
Windows version: 10.0.26200.8037
PS C:\windows\system32>
```

If WSL is already installed, update it using **wsl --update**

Install Docker Desktop

Visit <https://docs.docker.com/desktop/setup/install/windows-install/> to install Docker.

The image shows a screenshot of the Docker Docs website and the Docker Desktop installer configuration window.

Docker Docs Website:

- Header:** dockrdocs, Get started, Guides, Manuals, Reference, Gordon, Search, and a Docker logo.
- Left Sidebar:**
 - AI AND AGENTS:** Overview, Docker Sandboxes, MCP Catalog and Toolkit (Beta), Model Runner, Gordon, AI and Docker Compose, Docker Agent.
 - APPLICATION DEVELOPMENT:** Docker Desktop, Setup, Install, Mac, Mac permission requirements, Windows (highlighted), Windows permission requirements, Linux, VM or VDI environments, Sign in, Allowlist, Explore Docker Desktop.
- Main Content:**
 - ## Install Docker Desktop on Windows
 - Docker Desktop terms:** Commercial use of Docker Desktop in larger enterprises (more than 250 employees OR more than \$10 million USD in annual revenue) requires a paid subscription.
 - Table of contents:** Installation modes, System requirements, Install Docker Desktop on Windows, Install interactively, Install from the command line, Start Docker Desktop, Advanced system configuration, WSL: Verification and setup, Installer flags, Administrator privileges, Windows containers, Where to go next.
 - Buttons:** Docker Desktop for Windows - x86_64, Docker Desktop for Windows - x86_64 on the Microsoft Store, Docker Desktop for Windows - Arm (Early Access), Give feedback.

Docker Desktop Installer Configuration Window:

Name	Date modified	Type	Size
▼ Today			
Docker Desktop Installer.exe	5/22/2026 6:22 AM	Application	640,256 KB

Configuration Window:

Configuration

- ☒ All users installation (requires password)
 - ☒ Use WSL 2 instead of Hyper-V
 - ☐ Allow Windows Containers
 - Windows Containers should only be enabled if you understand the risks. For more information, see [our docs](#).
- ☐ Per-user installation (Beta)

☒ Add shortcut to desktop

Buttons: Cancel, OK

Docker Desktop 4.74.0

Installation succeeded

You must restart Windows to complete installation.

Close and restart

Check installed WSL distributions by `wsl --list --verbose`. If you only see something like this, then you do not yet have a normal WSL Linux distro installed. In other words, docker-desktop is not the Linux environment we normally need for our workshop.

```
Administrator: Windows PowerShell
PS C:\windows\system32> wsl --list --verbose
  NAME                STATE      VERSION
* docker-desktop      Running    2
PS C:\windows\system32>
```

You need to install ubuntu by running `wsl --install -d Ubuntu`.

```
igarousi@igarousidell: /mnt/c/windows/system32
PS C:\windows\system32> wsl --install -d Ubuntu
Downloading: Ubuntu
Installing: Ubuntu
Distribution successfully installed. It can be launched via 'wsl.exe -d Ubuntu'
Launching Ubuntu...
Provisioning the new WSL instance Ubuntu
This might take a while...
Create a default Unix user account: igarousi
New password:
Retype new password:
passwd: password updated successfully
usermod: no changes
Help improve Ubuntu!

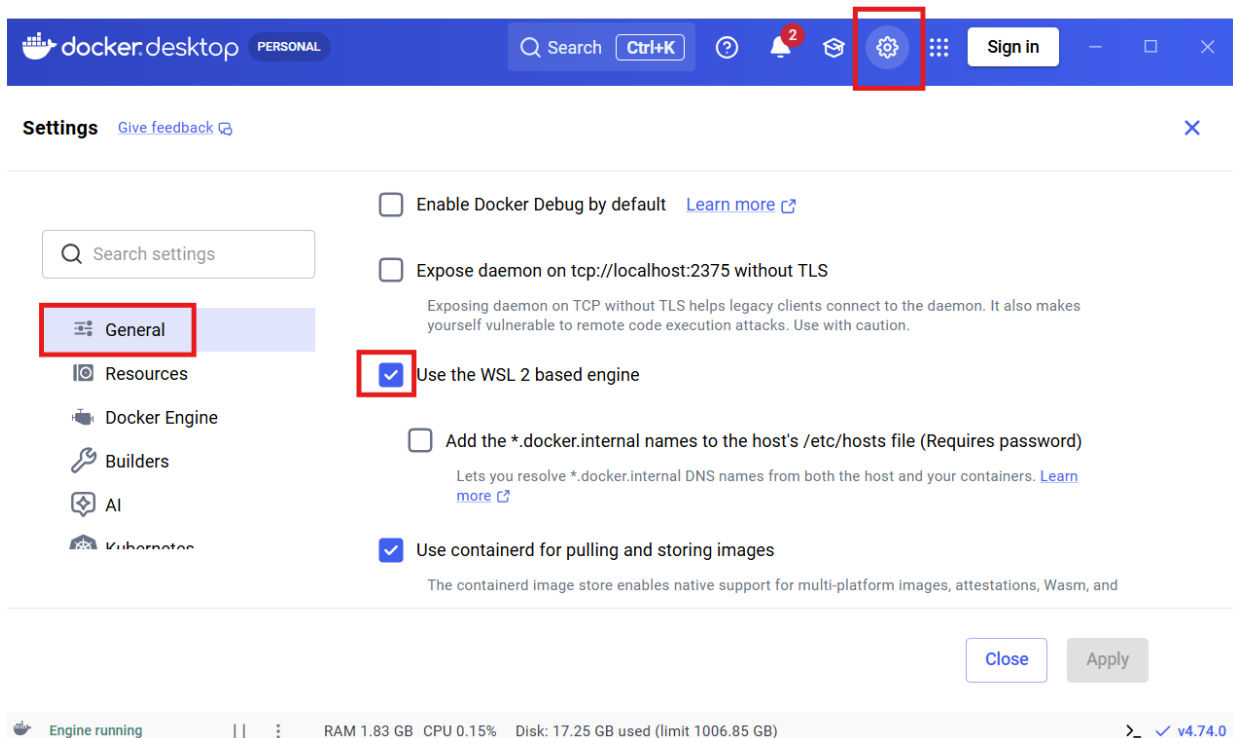
Help us improve Ubuntu features and compatibility by sharing system reports with Canonical.
Reports are sent anonymously and do not contain any personal data.
For legal details, please visit: https://ubuntu.com/legal/systems-information-notice

We will save your answer to Windows and will only ask you once.

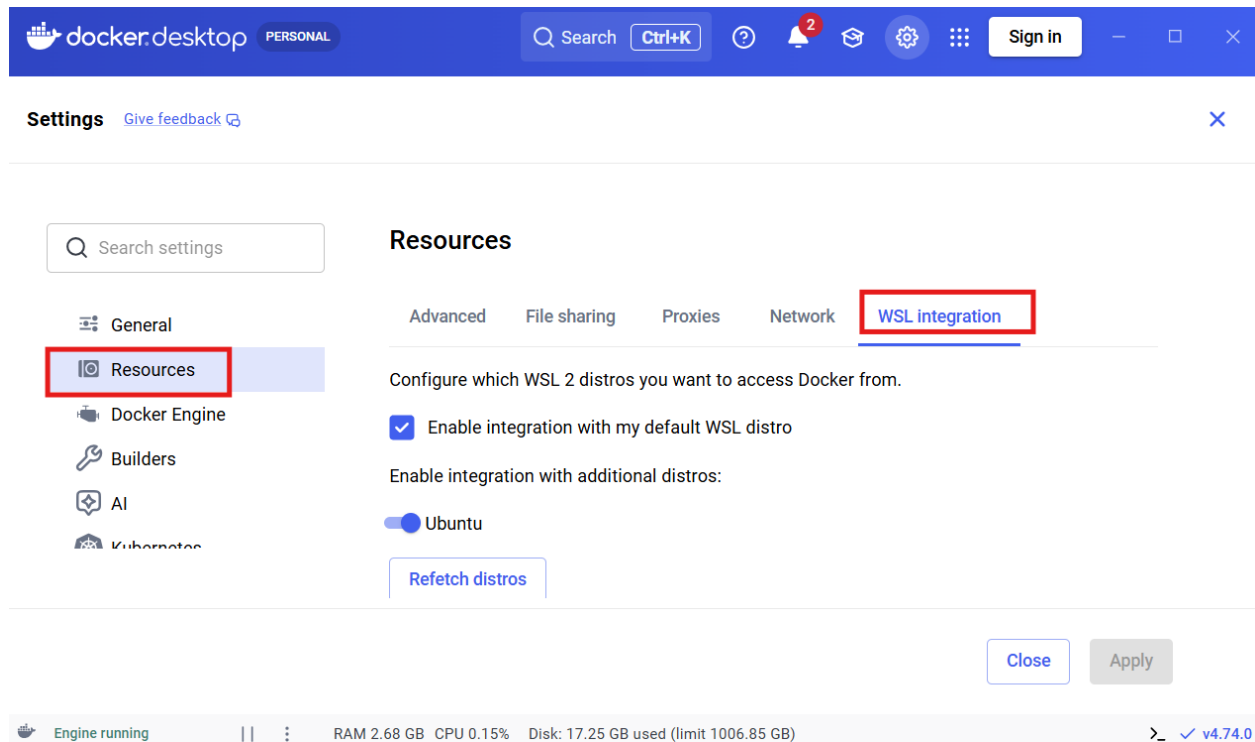
Would you like to opt-in to platform metrics collection (Y/n)? To see an example of the data collected, enter 'e'.
[Y/n/e]: n
igarousi@igarousidell: /mnt/c/windows/system32$
```

```
Windows PowerShell
PS C:\windows\system32> wsl --list --verbose
NAME                STATE      VERSION
* docker-desktop    Running    2
Ubuntu              Running    2
PS C:\windows\system32> |
```

Open Docker Desktop, got to Settings, select General, and make sure 'Use the WSL 2 based engine' is checked. Click Apply and restart.



Next, add Docker Desktop WSL integration by checking 'Enable integration with my default WSL distro' as well as 'Ubuntu'. Without this, Docker exists but fails inside WSL.



Install Python

Visit <https://www.python.org/downloads/windows/> and install the stable release.

Install Git

Visit https://git-scm.com/install/windows?utm_source and install the latest version for Windows.

Create A HydroShare Account

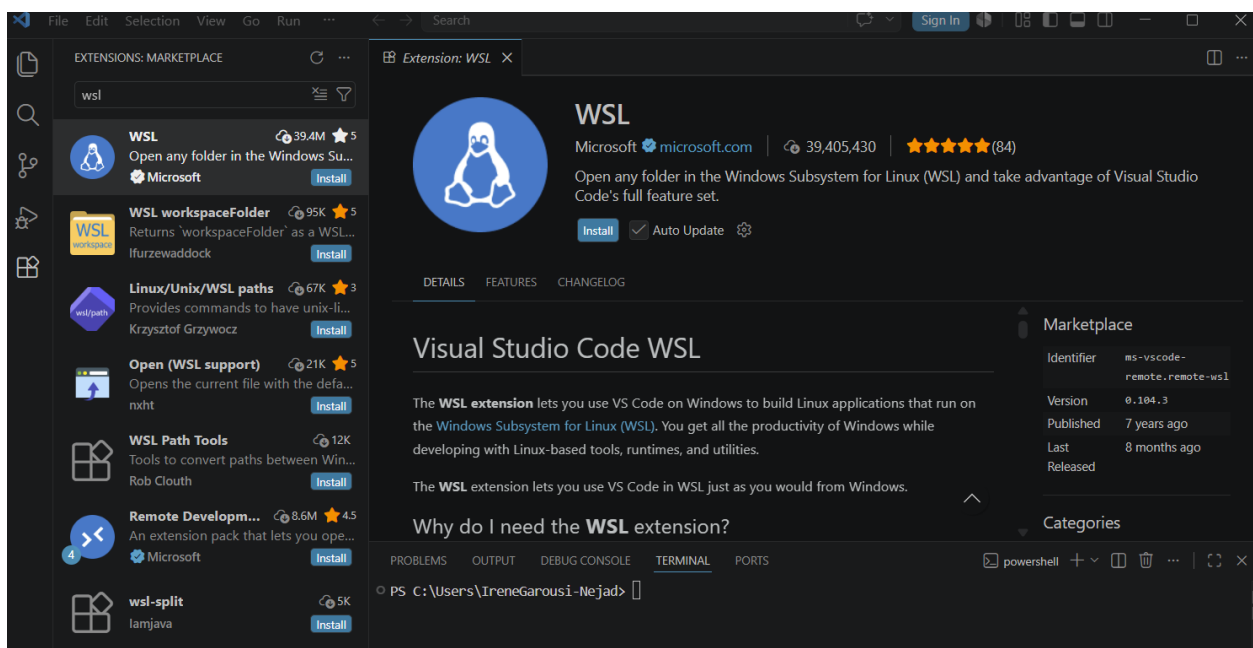
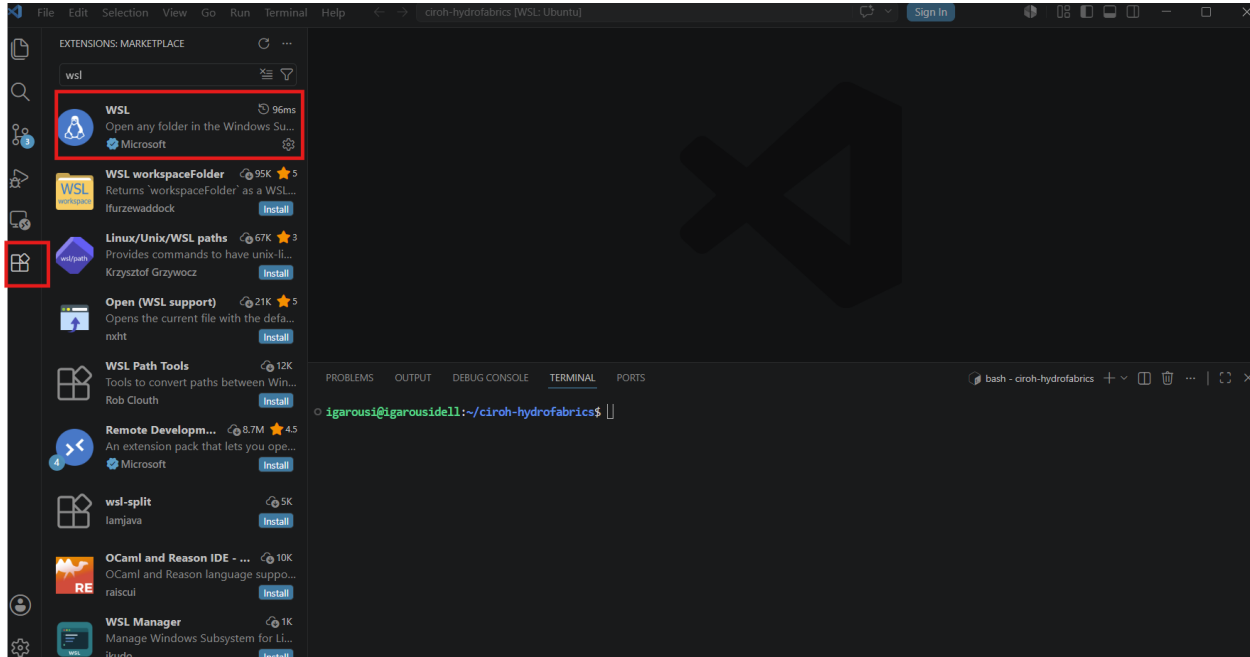
Visit <https://help.hydroshare.org/introduction-to-hydroshare/getting-started/create-an-account/> and create a free HydroShare account if you don't have one already.

Install VS Code

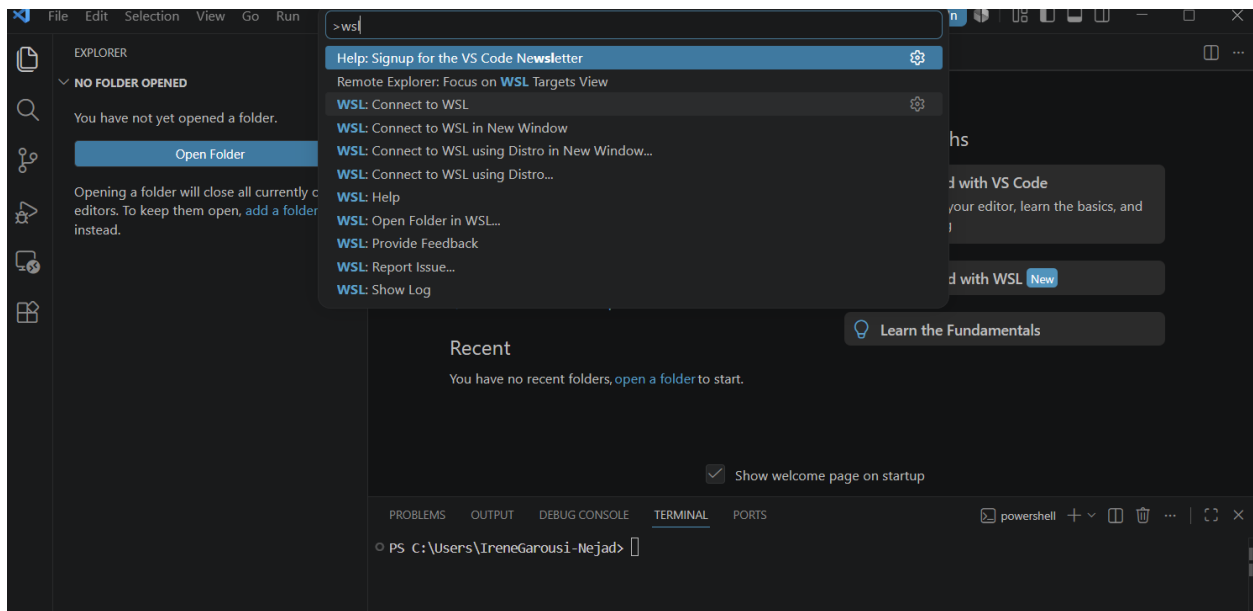
Visit <https://code.visualstudio.com/download> and install VS code.

In what follows, our goal is to connect VS Code to Ubuntu to allow having WSL terminal in VS Code.

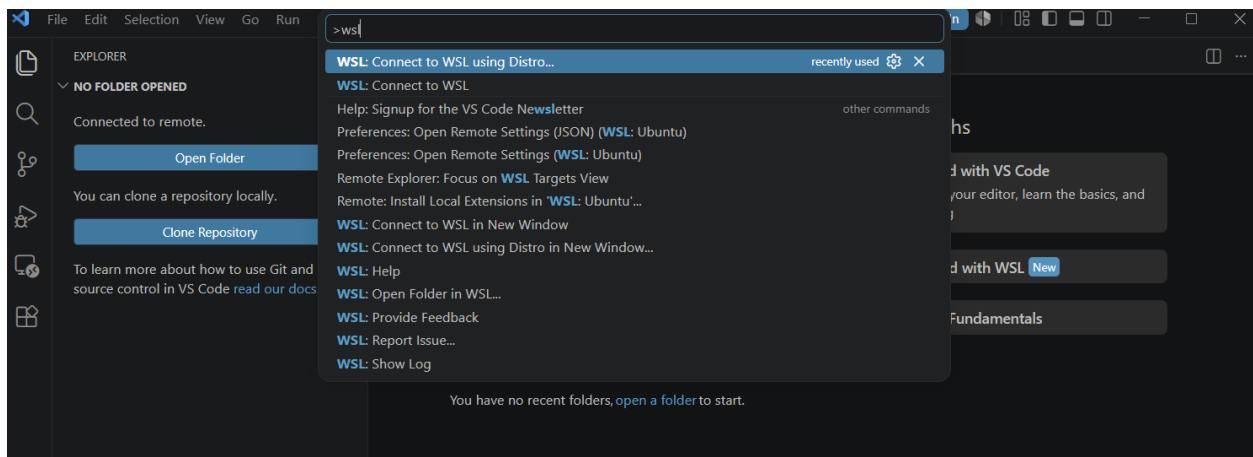
Open VS Code and install WSL extension.



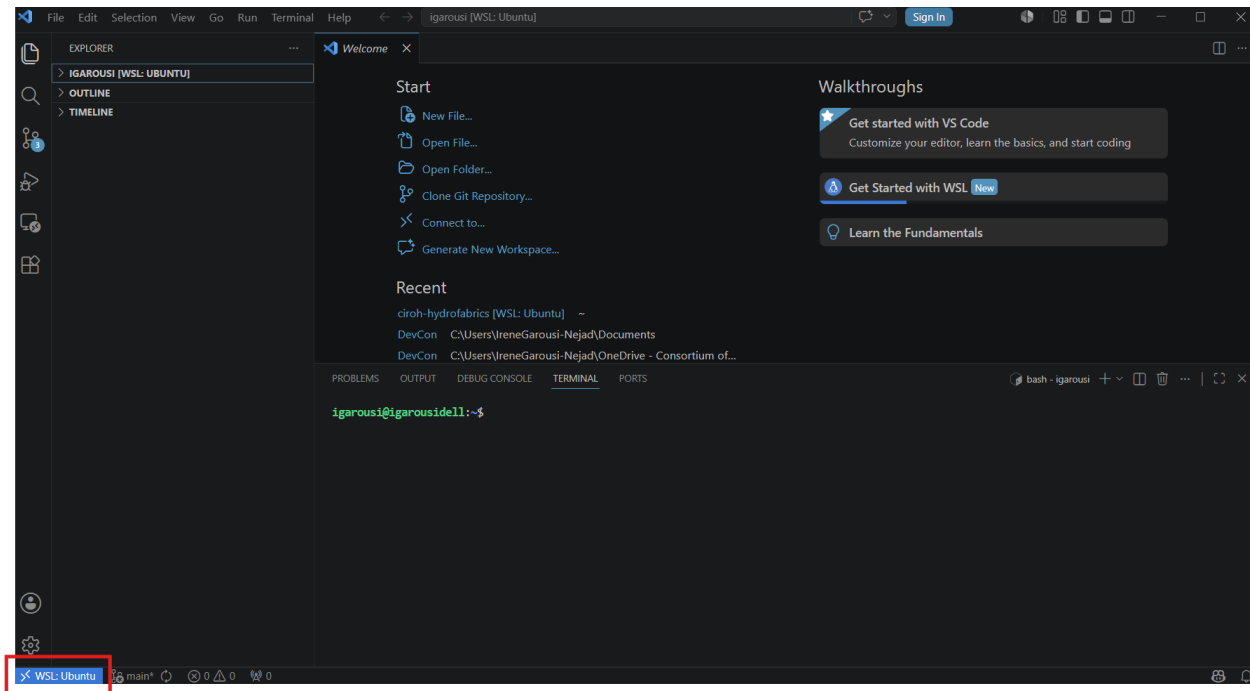
After installing WSL extension, open WSL windows using 'Ctrl + shift + P' and search for wsl.



Click on 'WSL: Connect to WSL using Distro' and then choose 'Ubuntu'.



Then you should see 'WSL: Ubuntu' is selected at the left corner.



Allow Docker access in WSL by running the following and then restart WSL and reopen it:

```
sudo usermod -aG docker $USER
```

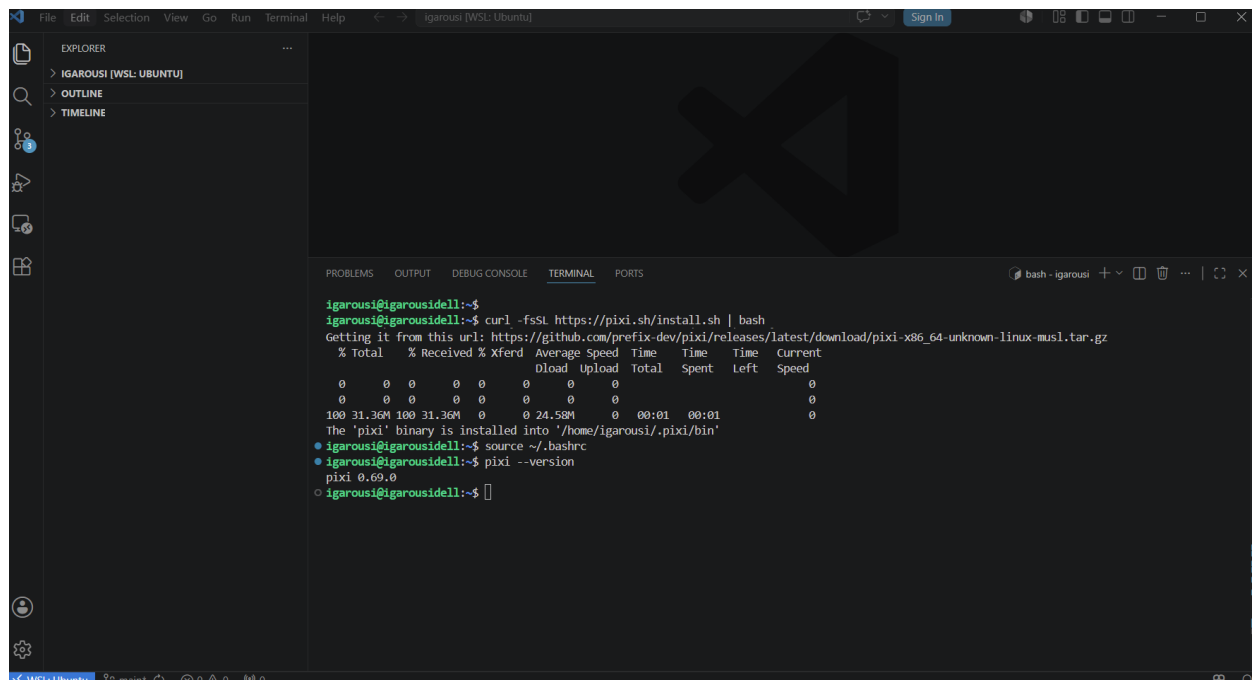
Install Pixi

Open Ubuntu WSL terminal within VS Code and run:

```
curl -fsSL https://pixi.sh/install.sh | bash
```

```
source ~/.bashrc
```

```
pixi --version
```



```
igarousi@igarousidell:~$ curl -fsSL https://pixi.sh/install.sh | bash
Getting it from this url: https://github.com/prefix-dev/pixi/releases/latest/download/pixi-x86_64-unknown-linux-musl.tar.gz
% Total % Received % Xferd Average Speed Time Time Time Current
         Dload Upload Total Spent Left Speed
  0     0     0      0      0      0      0      0      0
100 31.36M 100 31.36M      0  24.58M      0  00:01  00:01
The 'pixi' binary is installed into '/home/igarousi/.pixi/bin'
igarousi@igarousidell:~$ source ~/.bashrc
igarousi@igarousidell:~$ pixi --version
pixi 0.69.0
igarousi@igarousidell:~$
```

Additional Pixi workspace configuration will be done later during the workshop after cloning the repository.

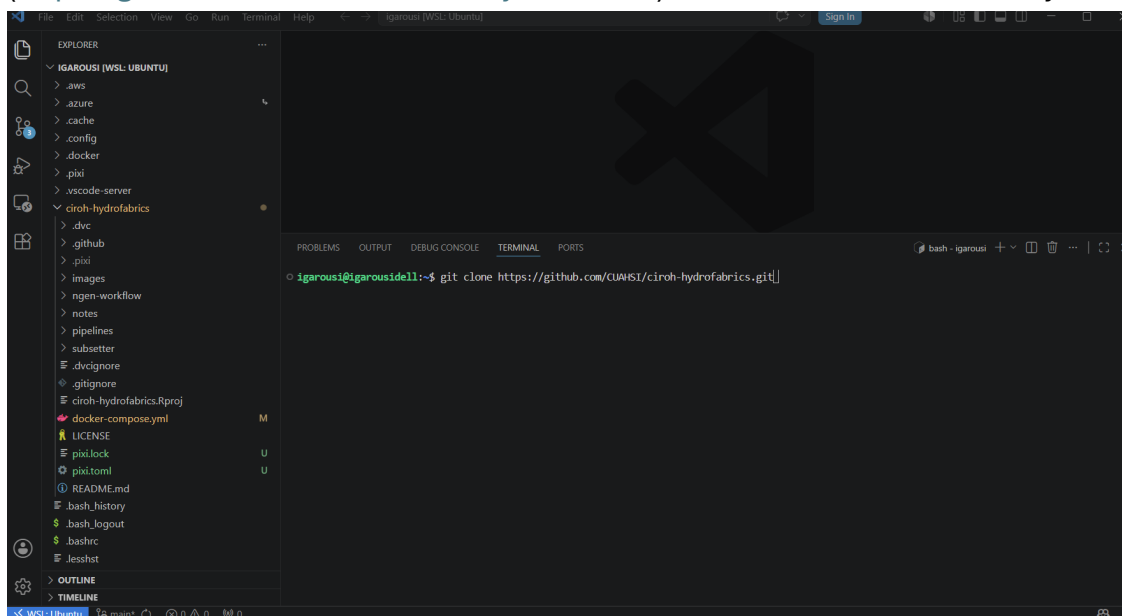
Install the QGIS

Visit <https://qgis.org/download/> and download the latest release. This is optional. You are welcome to use any GIS tool of your choice to view results of the workflow.

4. During the Workshop: Guided Workflow Steps

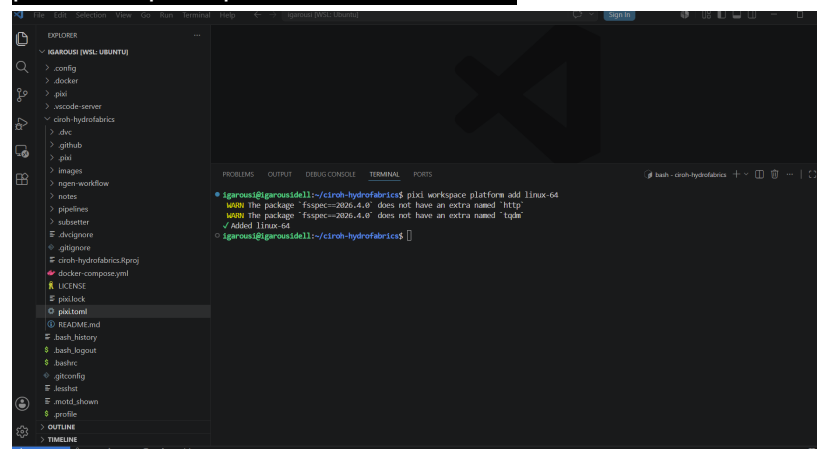
The following steps will be completed together during the workshop. They are shared in advance so you can become familiar with the workflow, expected commands, and overall process.

1. Overview of the reference and derived hydrofabrics
2. Introduce how Data Version Control connects GitHub, data, and workflow outputs
3. Run Docker Desktop
4. Open WSL Ubuntu terminal in VS Code and clone the GitHub repository
(<https://github.com/CUAHSI/ciroh-hydrofabrics>) **INSIDE** WSL home directory.



5. Prepare the Pixi environment. Go to the '**ciroh-hydrofabrics**' folder and run the following command to add required Pixi platform fix.

- **pixi workspace platform add linux-64**



- Then, run the following to download packages and create the environment.

pixi install

```

igarousi@igarousidell:~/ciroh-hydrofabrics$ pixi install
✓ The default environment has been installed.
igarousi@igarousidell:~/ciroh-hydrofabrics$

```

- Activate the environment by running **'pixi shell'** and then confirm the required **'aws'** and **'dvc'** packages are properly installed in this pixi environment:

```

igarousi@igarousidell:~/ciroh-hydrofabrics$ pixi shell
(codes) igarousi@igarousidell:~/ciroh-hydrofabrics$ aws --version
aws-cli/1.45.0 Python/3.11.15 Linux/6.6.114.1-microsoft-standard-WSL2 botocore/1.43.0
(codes) igarousi@igarousidell:~/ciroh-hydrofabrics$ dvc --version
3.67.1
(codes) igarousi@igarousidell:~/ciroh-hydrofabrics$

```

- Open the **'docker-compose.yml'** file and specify the correct docker image:

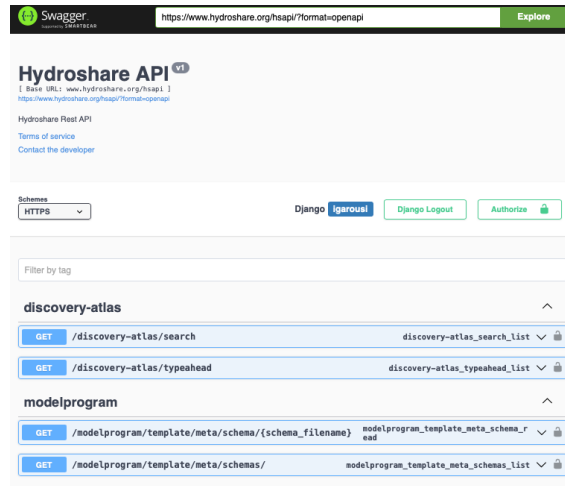
```

1 services:
2   hydrofabric:
3     platform: linux/amd64
4     user: "1000:1000"
5     image: cuahsi/ciroh-hydrofabrics-hydrofabric:latest # uncomment to use pre-built image from DockerHub
6     build:
7       context: ./ngen-workflow/
8       dockerfile: Dockerfile
9     volumes:
10      - ./ngen-workflow:/ngen-workflow
11     working_dir: /ngen-workflow
12     ulimits:
13       stack: 67108864
14

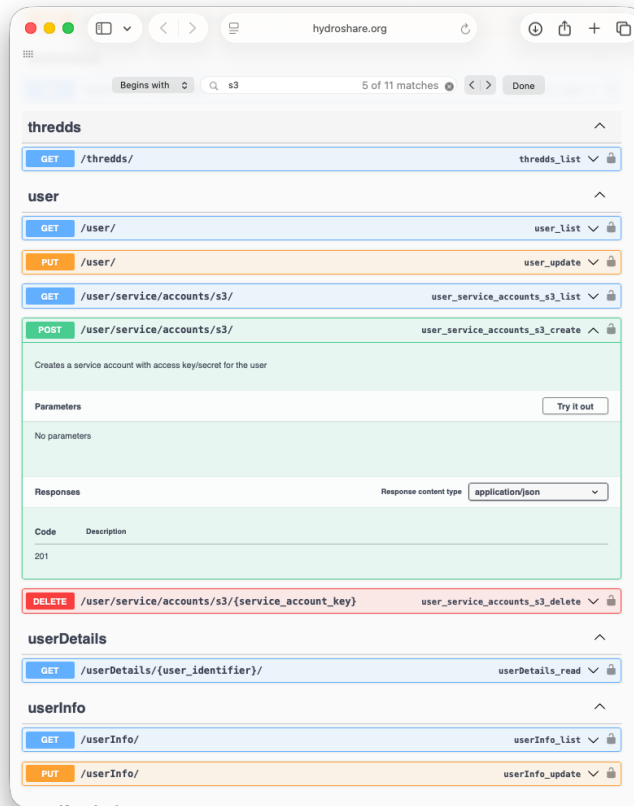
```

6. HydroShare S3 Authentication for DVC

- DVC needs access to the HydroShare-backed S3 storage where the workflow input files and cached outputs are stored. To enable this access, you need to follow these steps to configure an AWS credentials profile name **'hydrosahre'**:
 - Sign in to your HydroShare account at <https://www.hydroshare.org/>
 - Visit <https://www.hydroshare.org/hsapi/>



- Search for 'POST /user/service/account/s3/'



- Try it out and execute

POST /user/service/accounts/s3/ user_service_accounts_s3_create

Creates a service account with access key/secret for the user

Parameters Cancel

No parameters

Execute

Responses Response content type: application/json

Code	Description
201	

- Copy **'access_key'** and **'secret_key'**

Responses Response content type: application/json

Curl

```
curl -X 'POST' \
  'https://www.hydroshare.org/hsapi/user/service/accounts/s3/' \
  -H 'accept: application/json' \
  -H 'X-CSRF-TOKEN: zHh1KEZ7bWAK3WSzQZkMfwbcp1LND8G9Er6tghyWLIIEFmLGqWfUswxTBfwWq0' \
  -d ''
```

Request URL

https://www.hydroshare.org/hsapi/user/service/accounts/s3/

Server response

Code	Details
201	Response body <pre>{ "access_key": "YML16L1I "secret_key": "h0MnnpaK }</pre> Response headers <pre>access-control-allow-credentials: true access-control-allow-origin: https://www.hydroshare.org allow: GET,POST,HEAD,OPTIONS alt-svc: h3="443"; ma=2592000 content-language: en content-length: 93 content-type: application/json date: Thu, 21 May 2026 13:31:43 GMT server: gunicorn vary: Accept-Origin,Accept-Language,Cookie via: 1.1 google</pre> Request duration 1616 ms

Responses

Code	Description
201	

- In your terminal, run:


```
aws configure --profile hydroshare
```

```
aws configure set profile.hydroshare.endpoint_url https://s3.hydroshare.org
```
- When prompted, enter the HydroShare S3 credentials


```
AWS Access Key ID: <enter your access key>
```

```
AWS Secret Access Key: <enter your secret key>
```

```
Default region name: us-east-1
```

```
Default output format: json
```

- Inspect the project structure and key files, including workflow files, parameter files, and DVC-related metadata.
- Review the project structure and workflow components. We will walk through:
 - How the GitHub repository is organized and explain the role of the main folders and files including:
 - DVC-tracked data

- DVC remote data on HydroShare
- Workflow stages in dvc.yaml
- Workflow parameters in params.yaml
- Exact recorded outputs in dvc.lock

9. Reproduce an existing workflow state

- The goal is to demonstrate reproducibility: another researcher should be able to use the same code, data references, and same workflow definition to reproduce the same results on their own machine. We will run the workflow in a way that confirms the existing outputs can be **reconstructed** or pulled without needing to recompute everything.

```
dvc repro --pull pipelines/demo/devcon/dvc.yaml
```

```
git statu
```

```
git diff -- pipelines/demo/devcon/dvc.lock
```

10. Modify workflow parameters for the next exercise

- We will modify selected parameters in `params.yaml` that affect the resulting hydrofabric resolution. This shows how a researchers can use the same workflow to test a different configuration while keeping the process documented, tracked, and reproducible in the future.
- Visualize and compare outputs in QGIS

11. Contribute a hydrofabric correction

- In this example, we will demonstrate how a community-identified hydrofabric issue can be turned into a structured, reproducible, and transparent contribution. The goal is to show how local expert knowledge can be used to identify an issue, prepare a correction, track the associated data changes with DVC, and document the update in a way others can view and reproduce.
- Visualize and compare outputs in QGIS

12. Discuss how this supports community contribution.